

```

Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::a630:4f17:ba08:df22%7(Preferred)
IPv4 Address. . . . . : 192.168.1.103(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Friday, October 27, 2023 2:30:55 PM

```

Figure: Client's IP address

```

Pinging bux-home.bracu.ac.bd [103.69.149.102] with 32 bytes of data:
Reply from 103.69.149.102: bytes=32 time=3ms TTL=54
Reply from 103.69.149.102: bytes=32 time=3ms TTL=54
Reply from 103.69.149.102: bytes=32 time=3ms TTL=54
Reply from 103.69.149.102: bytes=32 time=2ms TTL=54

```

Figure: Server's IP address

HTTP REQUEST:

```

▼ Frame 13261: 489 bytes on wire (3912 bits), 489 bytes captured (3912 bits) on interface \Device\NPF_{46CF7187-F3AA-4E93-A2BE-215C6C249922}, id 0
  Section number: 1
  > Interface id: 0 (\Device\NPF_{46CF7187-F3AA-4E93-A2BE-215C6C249922})
  Encapsulation type: Ethernet (1)
  Arrival Time: Oct 27, 2023 19:26:35.008372000 Bangladesh Standard Time
  [Time shift for this packet: 0.000000000 seconds]
  Epoch Time: 1698413195.008372000 seconds
  [Time delta from previous captured frame: 0.000113000 seconds]
  [Time delta from previous displayed frame: 0.000000000 seconds]
  [Time since reference or first frame: 56.972737000 seconds]
  Frame Number: 13261
  Frame Length: 489 bytes (3912 bits)
  Capture Length: 489 bytes (3912 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ip:tcp:http]
  [Coloring Rule Name: HTTP]
  [Coloring Rule String: http || tcp.port == 80 || http2]

```

This is the Data Link Layer. This layer is responsible for moving frames from one hop to the next.

```

▼ Ethernet II, Src: Giga-Byt_00:bb:ad (18:c0:4d:00:bb:ad), Dst: TP-Link_b7:55:e1 (5c:62:8b:b7:55:e1)
  > Destination: TP-Link_b7:55:e1 (5c:62:8b:b7:55:e1)
  > Source: Giga-Byt_00:bb:ad (18:c0:4d:00:bb:ad)
  Type: IPv4 (0x0800)

```

Ethernet II falls under Data Link Layer.

Destination MAC address: 5c:62:8b:b7:55:e1 (Server)

Source MAC address : 18:c0:4d:00:bb:ad (Client)

```

▼ Internet Protocol Version 4, Src: 192.168.1.103, Dst: 103.69.149.102
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 767
    Identification: 0x743b (29755)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.103
    Destination Address: 103.69.149.102

```

This is the Network Layer.

Source IP address: 192.168.1.103 (client)

Destination IP address: 103.69.149.102 (server).

```

▼ Transmission Control Protocol, Src Port: 7754, Dst Port: 80, Seq: 1, Ack: 1, Len: 435
  Source Port: 7754
  Destination Port: 80
  [Stream index: 327]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 435]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 1234649816
  [Next Sequence Number: 436 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 2536655549
  0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x018 (PSH, ACK)
    Window: 1024
    [Calculated window size: 262144]
    [Window size scaling factor: 256]
    Checksum: 0xc088 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  > [Timestamps]
  > [SEQ/ACK analysis]
  TCP payload (435 bytes)

```

TCP falls under the Transport layer.

Source port: 7754 (Client)

Destination port: 80 (Server)

SEQ and ACK number 1, indicates the order of data transmission.

Flags: Push and Acknowledgement

Window size: 1024, means client can receive up to data size 1024 at a time.

```

Hypertext Transfer Protocol
> GET / HTTP/1.1\r\n
Host: bux-home.bracu.ac.bd\r\n
Connection: keep-alive\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36\r\n
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
Accept-Encoding: gzip, deflate\r\n
Accept-Language: en-US,en;q=0.9\r\n
> [truncated]Cookie: _fbp=fb.2.1696270795960.650819108; _ga_92NRF30D88=GS1.1.1697658397.3.0.1697658407.0.0.0; _ga=GA1.3.1306658748.1696270795; _ga_GW
\r\n
[Full request URI: http://bux-home.bracu.ac.bd/]
[HTTP request 1/1]
[Response in frame: 14173]

```

HTTP GET request was sent using HTTP v1.1

Host specifies the domain name

Keep-alive type connection is used

The accepted headers, languages, and encoding are also specified.

Previous cookie data is also sent with the request.

HTTP RESPONSE:

```

Frame 13263: 220 bytes on wire (1760 bits), 220 bytes captured (1760 bits) on interface \Device\NPF_{46CF7187-F3AA-4E93-A2BE-215C6C249922}, id 0
Section number: 1
> Interface id: 0 (\Device\NPF_{46CF7187-F3AA-4E93-A2BE-215C6C249922})
Encapsulation type: Ethernet (1)
Arrival Time: Oct 27, 2023 19:26:35.012687000 Bangladesh Standard Time
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1698413195.012687000 seconds
[Time delta from previous captured frame: 0.001021000 seconds]
[Time delta from previous displayed frame: 0.004315000 seconds]
[Time since reference or first frame: 56.977052000 seconds]
Frame Number: 13263
Frame Length: 220 bytes (1760 bits)
Capture Length: 220 bytes (1760 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp:http]
[Coloring Rule Name: HTTP]
[Coloring Rule String: http || tcp.port == 80 || http2]

```

This is the Data Link Layer. This layer is responsible for moving frames from one hop to the next.

```

Ethernet II, Src: TP-Link_b7:55:e1 (5c:62:8b:b7:55:e1), Dst: Giga-Byt_00:bb:ad (18:c0:4d:00:bb:ad)
> Destination: Giga-Byt_00:bb:ad (18:c0:4d:00:bb:ad)
> Source: TP-Link_b7:55:e1 (5c:62:8b:b7:55:e1)
Type: IPv4 (0x0800)

```

Ethernet II falls under Data Link Layer.

Destination MAC address: 18:c0:4d:00:bb:ad (Client)

Source MAC address : 5c:62:8b:b7:55:e1 (Server)

```

✓ Internet Protocol Version 4, Src: 103.69.149.102, Dst: 192.168.1.103
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 206
    Identification: 0x1c17 (7191)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 53
    Protocol: TCP (6)
    Header Checksum: 0x6a58 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 103.69.149.102
    Destination Address: 192.168.1.103

```

This is the Network Layer.

Source IP address: 103.69.149.102 (server)

Destination IP address: 192.168.1.103 (client)

```

✓ Transmission Control Protocol, Src Port: 80, Dst Port: 7754, Seq: 1, Ack: 436, Len: 166
  Source Port: 80
  Destination Port: 7754
  [Stream index: 327]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 166]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 2536655549
  [Next Sequence Number: 167 (relative sequence number)]
  Acknowledgment Number: 436 (relative ack number)
  Acknowledgment number (raw): 1234650251
  0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x018 (PSH, ACK)
  Window: 501
  [Calculated window size: 64128]
  [Window size scaling factor: 128]
  Checksum: 0x30a8 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > [Timestamps]
  > [SEQ/ACK analysis]
  TCP payload (166 bytes)

```

TCP falls under the Transport layer.

Source port: 80 (Server)

Destination port: 7754 (Client)

SEQ no. 1 ; ACK no. 436

Flags: Push and Acknowledgement

Window size: 501, means server can receive up to data size 501 at a time.


```
▼ Hypertext Transfer Protocol
  > HTTP/1.1 308 Permanent Redirect\r\n
    Connection: close\r\n
    Location: https://bux-home.bracu.ac.bd/>\r\n
    Server: Caddy\r\n
    Date: Fri, 27 Oct 2023 13:26:35 GMT\r\n
  > Content-Length: 0\r\n
    \r\n
    [HTTP response 1/1]
    [Time since request: 0.004315000 seconds]
    [Request in frame: 13261]
    [Request URI: http://bux-home.bracu.ac.bd/]
```

This is HTTP Response

Version: HTTP/1.1

Status Code 308 means the client should change to new location for all future requests

Connection will be closed after this response